

EEP2 – Week 4

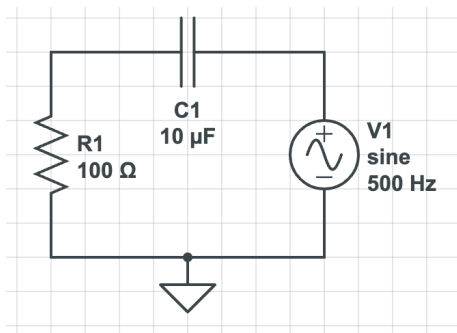
4/2/2025 – E4A-06-03

A0287717M – Brians Tjipto Meidianto

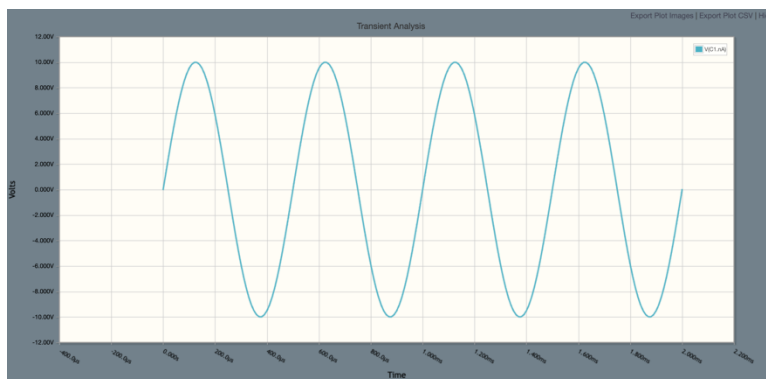
Tutorial

Exercise #1

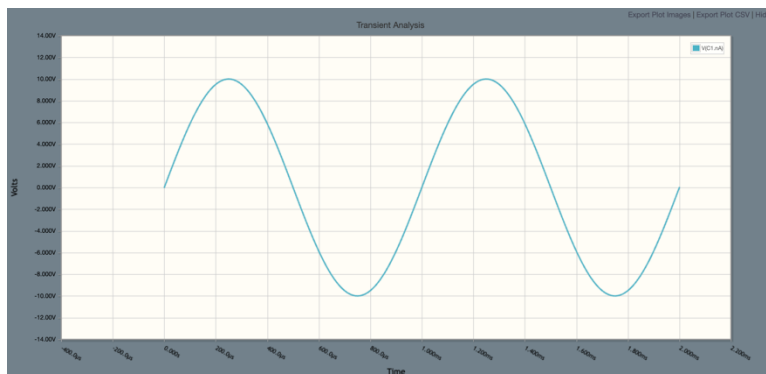
Capacitor:



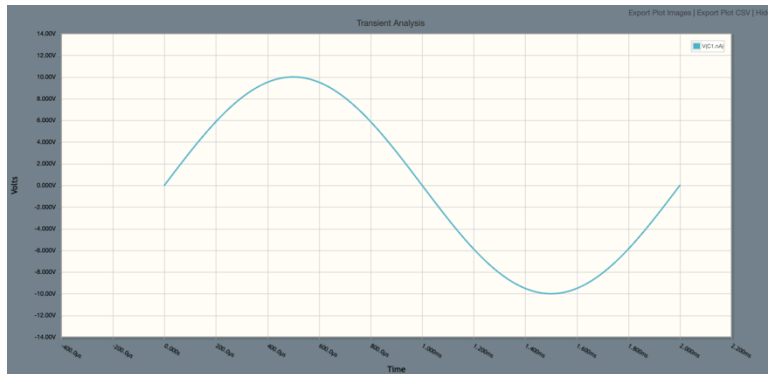
2kHz



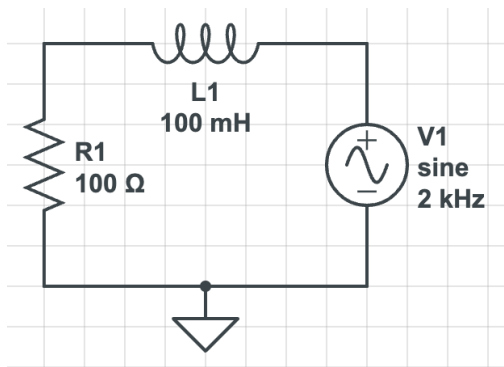
1kHz



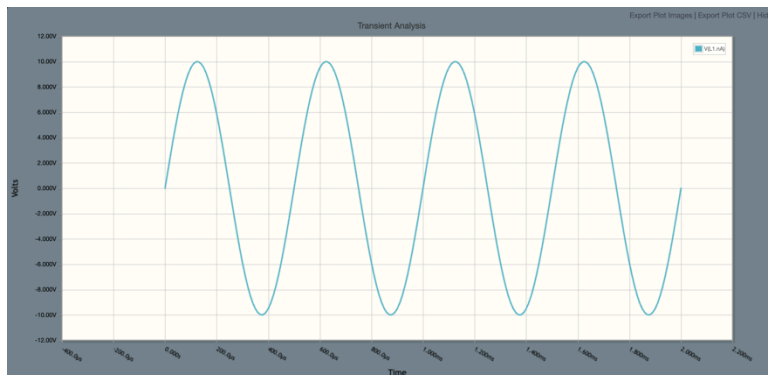
500Hz



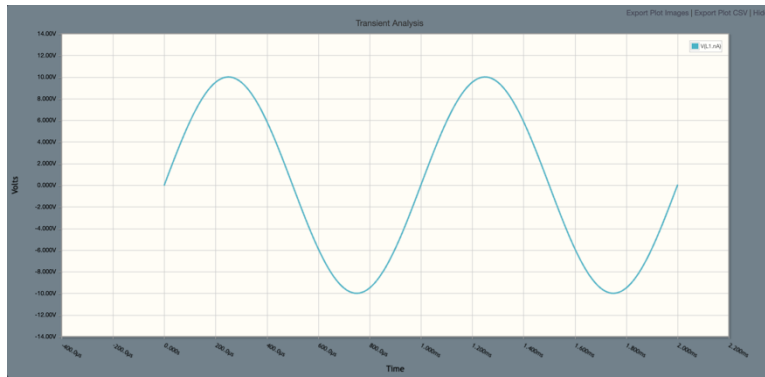
Inductor:



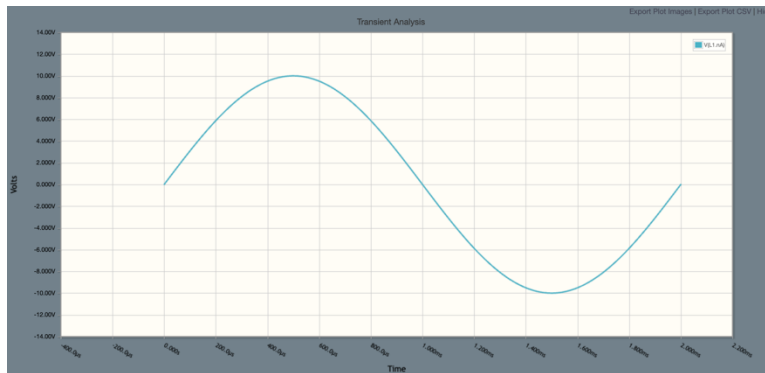
2kHz



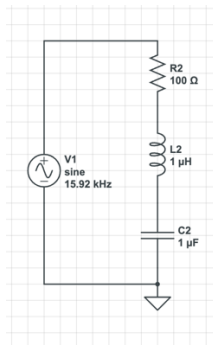
1kHz



500Hz

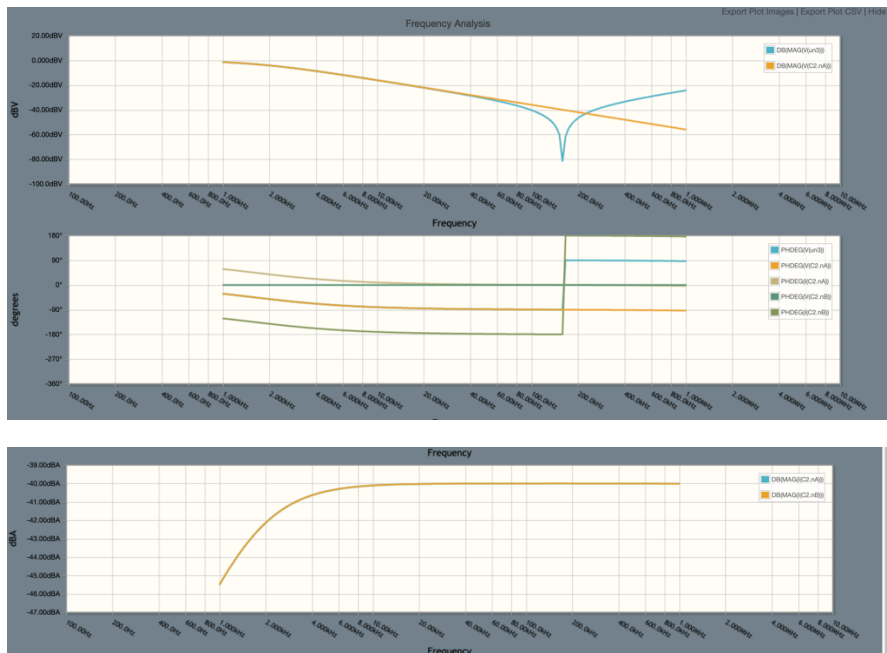


Exercise #2

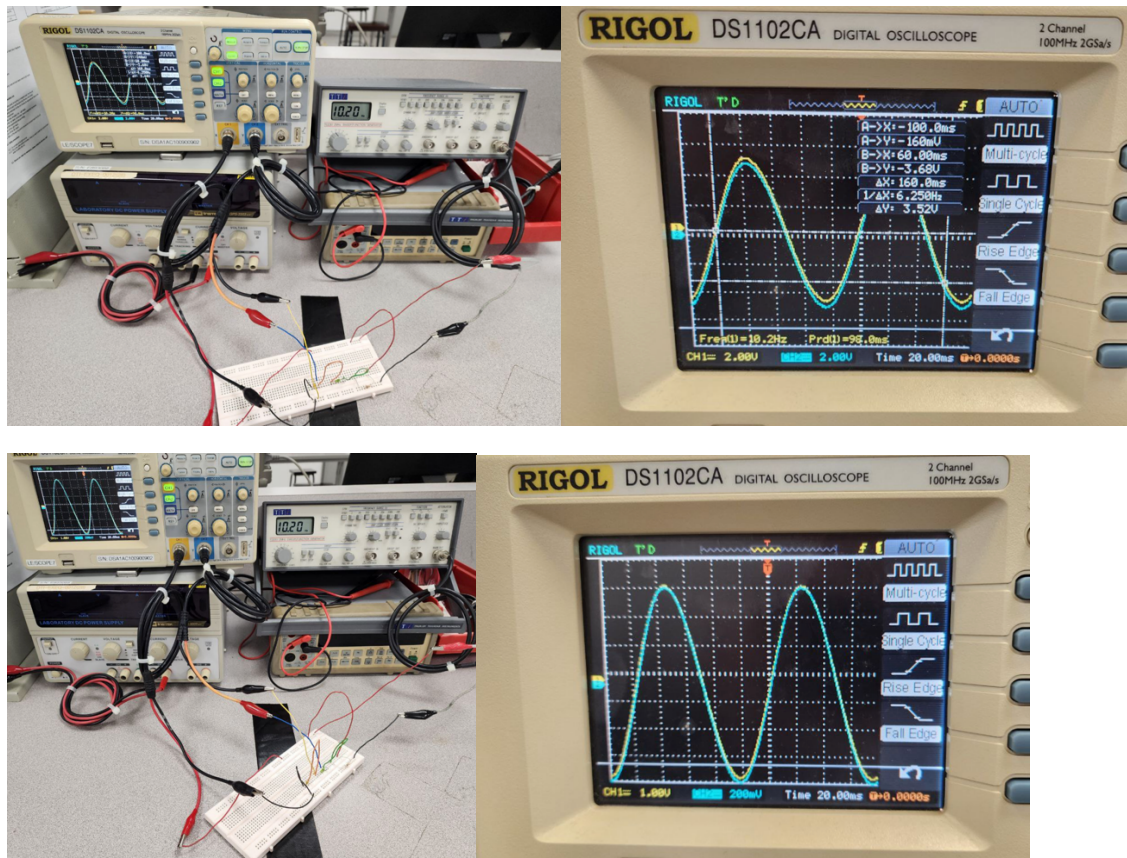


$$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(1 \times 10^{-3})(0.1 \times 10^{-6})}} = 100,000$$

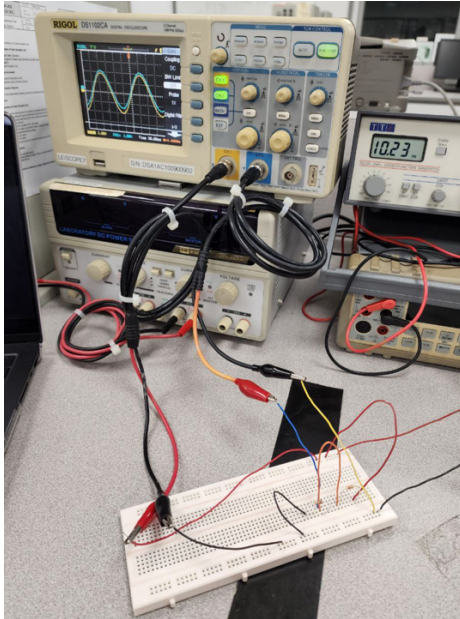
$$f_r = \frac{\omega_r}{2\pi} = \frac{10000}{2\pi} = 15915.5Hz = 15.92kHz$$



Lab



Activity #1



	R-R Circuit	L-R Circuit	C-R Circuit
Amplitude of V_s	4.99V	7.14V	5.76V
Amplitude of $V_{100\Omega}$	4.99V	4.99V	4.99V
Amplitude of $i(t)$	22.6mA	71.4mA	57.6mA
Phase of i w.r.t. V_s	0	0.342rad	-0.948rad
Current leads or lags	Neither	Lags	Leads

Activity #2

Theoretical Value:

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(1 \times 10^{-3}H)(1 \times 10^{-7}F)}} = 15.92kHz$$

$$\omega_r = 2\pi f_r = 2\pi \times 15.92 \times 10^3 = 100,000rad/s$$

